6 UAT (User Acceptance Testing)

User Acceptance Testing was performed by comparing the **original input image** with the **output images generated for each selected filter**. The purpose of UAT is to verify whether the system works correctly from the user's point of view and whether the generated outputs match the expected filter behavior.

6.1 UAT Test 1: Increase Brightness Operation

- **Input:** Original parrot image
- **Action:** User selects **Increase Brightness** in Python
- **Expected Output:** Image brightness should increase clearly
- **Observed Output:** Output image appears brighter than the original image
- **Status:** Accepted

6.2 UAT Test 2 : Decrease Brightness Operation

- Input: Original parrot image
- Action: User selects Decrease Brightness in Python
- Expected Output: Image brightness should reduce clearly
- Observed Output: Output image appears darker than the original image
- Status: Accepted

6.3 UAT Test 2 : Invert Operation

- Input: Original parrot image
- Action: User selects Invert Operation in Python
- Expected Output: Image should appear as a negative image
- Observed Output: Bright and dark regions are reversed in the output image
- Status: Accepted

6.4 UAT Test 2 : Threshold Operation

- Input: Original parrot image
- Action: User selects Threshold Operation in Python
- Expected Output: Image should convert into black and white format
- Observed Output: Output image shows binary segmentation based on intensity
- Status: Accepted

6.5 UAT Test 2 : Sobel Edge Detection

- Input: Original parrot image
- Action: User selects Sobel Edge Detection in Python
- Expected Output: Object boundaries and edges should be highlighted
- Observed Output: Edge regions of the parrot and background are detected
- Status: Accepted

6.6 UAT Test 2 : Gaussian Blur

- **Input:** Original parrot image
- **Action:** User selects **Gaussian Blur** in Python
- **Expected Output:** Image should appear smoother with reduced sharpness
- **Observed Output:** Output image shows smoothing effect compared to the original
- **Status:** Accepted

6.7 UAT Test 2 : Canny-style Edge Detection

- Input: Original parrot image
- Action: User selects Canny-style Edge Detection in Python
- Expected Output: Strong and weak edges should be highlighted
- Observed Output: Output image shows clear edge regions with reduced background detail
- Status: Accepted